

Valeria Criollo

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PROFESSIONAL SUMMARY

Immunologist (M.S., Comparative Biomedical Sciences) with 7+ years designing and running preclinical and immunopathology research, from in vivo model development through vaccine efficacy testing and publication. First-author of peer-reviewed studies in *Avian Diseases* and *Veterinary Immunology and Immunopathology*; skilled in flow cytometry, immune profiling, and advanced biostatistics (Prism, SAS, R). Co-founder of a regenerative-medicine company with an FDA-registered product and a filed patent. Native-bilingual English/Spanish.

TECHNICAL SKILLS

- **Immunology & assays:** Flow cytometry (multi-color panels) & FlowJo; ELISA; ELISpot; Western blot; qPCR/RT-qPCR; immunohistochemistry (IHC) & histopathology; immunopathology scoring; immune profiling
- **In vivo & cell biology:** In vivo model design & dosing; disease-challenge models; primary cell isolation; cell culture; necropsy; aseptic technique
- **Vaccine R&D:** Bacterial, viral & parasitic vaccine-candidate development; immunogenicity & protection testing
- **Biostatistics & data:** Experimental design; power/sample-size analysis; ANOVA, regression, survival analysis, mixed models; GraphPad Prism, SAS, R, Python, JMP
- **Rigor & compliance:** GLP; IACUC protocol development & compliance; SOP development; assay optimization & validation; scientific writing
- **Languages:** English (native/bilingual), Spanish (native/bilingual), Portuguese (conversational)

PROFESSIONAL & RESEARCH EXPERIENCE

North Carolina State University, College of Veterinary Medicine — Laboratory Manager

Raleigh, NC · Jan 2019 – Mar 2026

- Led immunopathology and preclinical research operations, managing and training **3–4 technical staff** and running **3+ sponsored studies in parallel** under **~\$300K/year** in industry research funding.
- Designed and validated **5 in vivo disease-challenge models** (4 bacterial, 1 coccidial) that supported vaccine development from concept through early-stage efficacy testing.
- Directed development of bacterial, viral, and parasitic vaccine candidates; evaluated immunogenicity and protection in challenge studies (**n ~ 30–100**), yielding **two first-author** peer-reviewed publications.
- Built flow cytometry, ELISA, IHC/histopathology, and qPCR workflows to characterize host immune responses and immunopathology.
- Conducted advanced biostatistical analysis (Prism, SAS, R) — experimental design, power analysis, ANOVA/regression/mixed models — to establish vaccine efficacy and immune correlates.

GliaBeauty — Co-Founder & Chief Executive Officer

Raleigh, NC (laboratories in Mexico) · Sep 2023 – Dec 2025

- Co-founded a regenerative-medicine venture — **GliaBeauty** (cosmetic) and its affiliated pharmaceutical/clinical entity **CryoGen, LLC** — developing **stem cell- and exosome-based** skin and hair therapies; directed R&D from concept to a **FDA-registered product** with an assigned National Drug Code (NDC).
- Designed and oversaw **preclinical testing (including in vivo models), clinical-trial design, and data analysis** through the CryoGen clinical arm.
- **Co-inventor** on a filed U.S. patent for skin-treatment compositions and methods (see Patents).
- Built and led a **team of 9** across US and Mexico-based laboratories and operations.

CEVA Santé Animale — Program Coordinator

Remote / Latin America & Europe · Jan 2019 – Feb 2026

- Coordinated an **international veterinary certification program** delivered in partnership with **NC State College of Veterinary Medicine**, extending professional certification to veterinarians across **Latin America and Europe**.
- Served as **bilingual liaison** providing **translation and clinical assistance** to program participants across multiple countries.

Duke University — Laboratory Technician, Bohórquez Lab

Durham, NC · Apr 2016 – Apr 2017

- Conducted neuroscience and biomedical-engineering research on **gut-brain signaling pathways** (neuropod cells) in the Bohórquez Lab, supporting study execution, sample processing, and data collection.

Duke University Health System — Animal Care Technician

Durham, NC · 2018

- Supported preclinical studies under **IACUC compliance** — animal husbandry, welfare monitoring, and study/lab support.

Earlier: Quality Control Assistant, Trofi Tropenfruchtimport GmbH (Hamburg, Germany, 2015) — quality inspection and import/export compliance documentation.

SELECTED PUBLICATIONS

1. **Criollo V**, Gaghan C, John F, Orozco E, Thachil A, Crespo R, Kulkarni RR. Immune Response Evaluation in Commercial Turkeys Affected with Clostridial Dermatitis. *Avian Diseases*. 2023;67(1):80–88. **(first author)**
2. **Criollo V**, John FA, Gaghan C, Fletcher OJ, Thachil A, Crespo R, Kulkarni RR. Characterization of immune responses and immunopathology in turkeys experimentally infected with clostridial dermatitis-producing strains of *Clostridium septicum*. *Veterinary Immunology and Immunopathology*. 2024;269:110717. **(first author)**
3. John FA, **Criollo V**, Gaghan C, Armwood A, Holmes J, Thachil AJ, Crespo R, Kulkarni RR. Immunization of turkeys with *Clostridium septicum* alpha toxin-based recombinant subunit proteins can confer protection against experimental clostridial dermatitis. *PLOS ONE*. 2024;19(4):e0302555.
4. Boyett T, Crespo R, **Vinueza VC**, Gaghan C, Mohammed JP, Kulkarni RR. Enumeration and speciation of coccidia affecting turkeys using flow cytometry method. *Journal of Applied Poultry Research*. 2022;31(3):100270.
5. Pejuán W, **Criollo V**, Paz PE. Optimum Harvest Time of the Nile Tilapia (*Oreochromis niloticus*) in Honduras: A Two-Step OLS Procedure to Obtain the Growth Function. *Fisheries and Aquaculture Journal*. 2019;10(3):270.

PATENTS

- Compositions and Methods of Use for the Treatment of Skin. U.S. Patent Application **US 2023/0277439 A1** (patent pending) — **co-inventor**.

EDUCATION

- **M.S., Comparative Biomedical Sciences (Immunology concentration)** — North Carolina State University, College of Veterinary Medicine, 2025
- **B.S., Business Administration / Agribusiness Management** — EAP Zamorano (Escuela Agrícola Panamericana), Honduras, 2015 — **Valedictorian**

CERTIFICATIONS, HONORS & AFFILIATIONS

- CITI Program — Responsible Conduct of Research (RCR)
- Duke University — Animal Research Coordinator
- Gamma Sigma Delta — The Honor Society of Agriculture
- Honors: Agribusiness Management Valedictorian; Meritorious Thesis (Zamorano)